

Comparing Futures and Forwards

This assignment gives you the opportunity to work through the distinctions between futures and forwards and justify ways in which they can be interchanged and ways in which they cannot.

Assume that there is a constant daily risk-free rate.

Part A: Daily pricing

1. Suppose there is one day until expiration, the futures price is F_{n-1} , and you have a futures position X . On expiration, the price is F_n . What is the value of your daily gain or loss on the futures position on expiration?

$$X(F_n - F_{n-1})$$

2. Suppose there are two days until expiration, the futures price is F_{n-2} , and you have a futures position X_{n-2} . At settlement on $n - 1$, what is the value of your gain or loss? What is value of that gain or loss at expiration?

At $n - 1$, the gain or loss is $X_{n-2}(F_{n-1} - F_{n-2})$. That compounds at the daily risk free rate for one day. Let's call the daily compounding factor R so that at expiration the gain or loss is

$$X_{n-2}(F_{n-1} - F_{n-2})R$$

3. Repeat step 2 for $n - 3, \dots, 1$, to 0. That is, what is the one-day gain or loss, and then what is its value compounded to expiration?

One day gain/loss:

$$X_{n-3}(F_{n-2} - F_{n-3})$$

$$X_1(F_2 - F_1)$$

$$X_0(F_1 - F_0)$$

Compounding to expiration:

$$X_{n-3}(F_{n-2} - F_{n-3})R^2$$

$$X_1(F_2 - F_1)R^{n-2}$$

$$X_0(F_1 - F_0)R^{n-1}$$

4. If there are n days until expiration, and the current forward price is FO_0 , what is value at expiration of a portfolio that invests the dollar amount FO_0 in risk-free bond and takes a long forward position Y_0 ? How can you select Y_0 to make the terminal value independent of FO_0 ?

Risk-free investment: $FO_0 R^n$

Forward position: $Y_0(S_T - FO_0)$

If we select the position $Y_0 = R^n$ (which assumes interest rate is a known constant), then the total portfolio is worth $S_T R^n$.

5. How can you select the positions $Y_0, X_0, X_1, \dots, X_{n-1}$ to make these values equal?

Select $Y_0 = R^n$ as in the previous part, and then select $X_i = R^{i+1}$ so that all the compounded gains and losses have the same factor R^n at expiration. The total gains and losses at expiration will cascade:

$$R^n(F_1 - F_0 + F_2 - F_1 + \dots F_n - F_{n-1}) = R^n(F_n - F_0)$$

The futures price at expiration will settle to $S_T = F_n$, so this is $R^n(S_T - F_0)$.

By investing F_0 at the risk free rate from 0 to n , the portfolio would be worth $R^n F_0 + R^n(S_T - F_0) = R^n S_T$, matching the terminal value of the forward portfolio. The initial investments F_0 and FO_0 have the same terminal value, therefore they must be equal by the usual no-arbitrage reasoning.

Part B: Portfolio value

Calculate the changing values of a futures position X_t compared to a forward position Y_t in the following steps.

1. For futures, you may assume all times fall on settlement. If you start with a futures position X_{t_1} on t_1 , and then rebalance to a position X_{t_2} at time t_2 , what is the value of the futures position and the gain/loss from rebalancing?

The futures position always resets to zero, so changing positions by $X_{t_2} - X_{t_1}$ costs nothing. The gains or losses will equal the initial position times the change in price:

$$X_{t_1}(F_{t_2} - F_{t_1})$$

In the limit, this is $X_t dF_t$

2. If you start with a forward position Y_{t_1} on t_1 , and then rebalance to a position Y_{t_2} at time t_2 , what is the value of the forward position and the gain/loss from rebalancing?

At t_2 , buy or sell $Y_{t_2} - Y_{t_1}$ forwards for zero change in the portfolio value at that time. The only change in value is from the change in the forward price for the initial position Y_{t_1} . At expiration, the S_T dependence will cancel, and the value of the Y_{t_1} entry/exit is $Y_{t_1}(FO_{t_2} - FO_{t_1})$. Discount back to t_2 to figure the value when the position is adjusted:

$$Y_{t_1} e^{-r(T-t_2)}(FO_{t_2} - FO_{t_1}).$$

Add and subtract the value discounted back to t_1 to see how the value has changed:

$$Y_{t_1} [e^{-r(T-t_1)}(FO_{t_2} - FO_{t_1}) + (e^{-r(T-t_2)} - e^{-r(T-t_1)})(FO_{t_2} - FO_{t_1})].$$

In the limit, this is

$$Y_t [e^{-r(T-t)} dFO_t - d(e^{-r(T-t)}) dFO_t]$$

The risk free rate is constant implying that the second term is zero:

$$Y_t e^{-r(T-t)} dFO_t.$$

3. Consider the continuous limit of these discrete formulas to find the relationship between the changing value of a forward position and a futures position. If you are hedging a position with 100 futures contracts, would the corresponding forward position be bigger or smaller?

If Y_t contracts are held in forwards, then $X_t = Y_t e^{-r(T-t)}$ should be held in futures. Under positive constant interest rates, the futures position is smaller than the forward position, so the forward position would be greater than the 100 futures contracts.